

Functional & Performance Testing

Model Performance Test

Date	01 July 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	

The Smart Agricultural Production Optimization Engine (OptiCrop) was tested after complete development and integration of the machine learning model, Flask backend, and web interface. Functional testing verified that all major system features operate correctly, while performance testing ensured acceptable response times and stable execution.

Functional Testing

Test Case ID	Scenario (What to Test)	Test Steps	Expected Result	Actual Result	Status
FT-01	Soil Parameter Input Validation	Enter valid and invalid values for N, P, K, Temperature, Humidity, pH, and Rainfall	Invalid inputs are rejected and valid inputs are accepted	Input validation worked correctly	Pass
FT-02	Crop Recommendation Prediction	Enter valid agricultural parameters and submit the prediction request	Recommended crop is displayed correctly	Crop recommendation generated successfully	Pass

FT-03	Crop Suitability Evaluation	Select a crop and provide environmental parameters	Suitability result is displayed	Suitability evaluation completed successfully	Pass
FT-04	Model Loading	Start the Flask application	Trained model (.pkl) loads without errors	Model loaded successfully during application startup	Pass
FT-05	API Endpoint Testing	Send prediction request to the Flask API	JSON response contains predicted crop	API returned valid prediction response	Pass
FT-06	Frontend Integration	Submit prediction through the web interface	Prediction displayed on the result page	Frontend successfully communicated with backend	Pass
FT-07	Dataset Analysis Module	Upload/load dataset and perform EDA	Statistics and visualizations generated	Dataset analysis completed successfully	Pass
FT-08	Invalid Request Handling	Submit incomplete or incorrect data	Error message displayed without application crash	Error handling worked correctly	Pass

Performance Testing

Test Case ID	Performance Test	Test Procedure	Expected Result	Actual Result	Status
PT-01	Prediction Response Time	Measure prediction time after submitting valid input	Less than 2 seconds	Average response time $\approx$ 0.4–0.8 seconds	Pass

PT-02	Model Loading Time	Measure application startup	Model loads successfully within a few seconds	Startup completed successfully	Pass
PT-03	Multiple Prediction Requests	Submit multiple prediction requests consecutively	Stable execution without failures	Application remained responsive	Pass
PT-04	Dataset Processing	Load agricultural dataset for analysis	Dataset processed successfully	Dataset loaded and analyzed correctly	Pass
PT-05	End-to-End Application Performance	Execute the complete workflow from user input to prediction	Successful execution without crashes	Entire workflow completed successfully	Pass

Machine Learning Model Performance

Metric	Result
Accuracy	0.9932 (or 99.32%)
Precision	0.9935
Recall	0.9932
F1-Score	0.9932
Best Performing Model	Random Forest

Summary

All functional test cases passed successfully. The integrated system correctly validated user inputs, generated crop recommendations, evaluated crop suitability, processed agricultural datasets, and maintained stable performance under repeated prediction requests. The Flask backend, machine learning model, and web interface worked together successfully, indicating that the Smart Agricultural Production Optimization Engine (OptiCrop) is functionally complete and ready for deployment and demonstration.

